Arithmetic profiles and the prime spectrum of the integer necklace ring

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Abstract. Let $R = Nr(\mathbb{Z})$ be the integer necklace ring and let $B = \prod_{n\geq 1} \mathbb{Z}$ be its ghost product. We study ideal radicals and the ordinary prime spectrum through this embedding. For every ideal $I \subseteq R$, we prove $(IB \cap R)^2 \subseteq I$. It follows that radicals contract and every prime of R lifts to B. We characterize the coordinatewise gcd profiles of finite ghost families by truncated valuation conditions and show that two ghosts realize every such profile. Consequently every finitely generated ideal of R has arithmetical rank at most two. The profiles give an arithmetic presentation of the lattice of quasicompact Zariski opens and of the spectrum, including specialization and the patch topology. A least compatible multiple construction gives infinitary divisibility criteria for equality and containment of ambient prime contractions. We also classify the primes closed in the necklace coordinate topology and describe all ambient primes above each of them. The resulting account distinguishes the full ordinary spectrum from its coordinate-topology-closed part and relates both to the known positive-characteristic and p-adic constructions.

Keywords. Necklace ring; big Witt vectors; prime spectrum; ideal contraction; arithmetical rank; ultrafilter.

1. Introduction

The integer necklace ring has an injective ghost map into a product of copies of $\mathbb{Z}$. In that product, multiplication and divisibility are coordinatewise. The image, however, is constrained by congruences linking different coordinates. These congruences make ideal theory in the necklace ring a separate problem: solving every coordinate equation does not necessarily solve an ideal-membership equation in the ring.

The failure already appears in the ideal generated by two. Write e_2 for the necklace vector with second coordinate one and all other coordinates zero. Its ghost coordinates are two at even indices and zero at odd indices. Thus $e_2 \in 2B \cap R$, but $e_2 \notin 2R$, since doubling a necklace vector doubles each of its integer coordinates. Nevertheless $e_2^2 = 2e_2 \in 2R$. The first result of this paper shows that the same repair by a product of two elements works for every ideal:

$$(IB \cap R)^2 \subseteq I \qquad (I \subseteq R). \tag{1.1}$$

This includes ideals that are neither finitely generated nor closed in the coordinate topology. Its proof constructs a fixed finite family of Bézout coefficients satisfying all ghost congruences. The construction is needed because separate coordinatewise choices of Bézout coefficients need not be ghosts.

Two consequences make (1.1) useful for studying the spectrum. First, $\sqrt{IB} \cap R = \sqrt{I}$ for every ideal I. Second, every prime of R is the contraction of a prime of B. This prime lifting does

not follow from the usual integral-extension theorem: we show that $R \subseteq B$ is not integral. For a finitely generated ideal $I = (a_1, \dots, a_r)$, radical membership becomes

$$b \in \sqrt{I} \quad \iff \quad \exists N \geq 1 \ \forall n \geq 1, \quad \gcd(g_n(a_1), \dots, g_n(a_r)) \mid g_n(b)^N. \tag{1.2}$$

The exponent must be independent of n. The uniformity in this condition retains arithmetic information that is lost by examining one coordinate at a time.

Our second main result identifies the integer sequences that occur as the gcd in (1.2). A sequence $d = (d_n)$ of nonnegative integers occurs if and only if, for every $p^k \| n$,

$$\min\{v_p(d_n), k\} = \min\{v_p(d_{n/p}), k\}, \qquad v_p(0) = \infty. \tag{1.3}$$

Moreover, two ghosts suffice to realize any sequence satisfying (1.3). Together with (1.1), this implies that every finitely generated ideal has the same radical as an ideal generated by two elements. Equivalently, every quasicompact Zariski open in $\operatorname{Spec} R$ is a union of two principal opens. The realization proof is constructive at successive integer coordinates; it does not assert that a gcd profile is itself a ghost.

These two results lead to a description of the entire ordinary spectrum. We form a lattice from the sequences satisfying (1.3), with its order expressed by uniform power divisibility. Standard spectral reconstruction then gives the points, specialization order and topology. We also give a direct point model by binary signatures, with a numerical test for every finite pattern. This is an exhaustive arithmetic presentation, but its tests are infinitary. It is not a finite enumeration or an algorithm deciding arbitrary prime membership.

The ambient product is useful for more explicit calculations. We construct a least compatible multiple of any nonnegative integer sequence and use it to decide which ambient ideals have included or equal contractions. For primes closed in the necklace coordinate topology, the resulting fibers can be described completely by ultrafilters and convex subgroups of ordered valuation groups. Here "closed" refers to closure of an ideal as a subset of the topological ring. It does not mean a closed point of the Zariski spectrum. The distinction is visible even in positive characteristic, where many maximal ideals are dense in the coordinate topology.

The necklace ring and its identifications with the big Witt ring and the completed Burnside ring of $\widehat{\mathbb{Z}}$ are established constructions. They originate in the work of Metropolis–Rota [1] and Dress–Siebeneicher [2], and are developed further in Oh's work on generalized Burnside and necklace rings [3–5]. Oh also treats decompositions of these functors [6], parameterized Witt–Burnside constructions [7] and their relation to λ-rings [8]. We use the integer specialization and keep the functorial identifications separate from the ideal-transfer assertions proved here. In particular, a decomposition by composition of Witt functors is not a product decomposition of the ring under study.

Several parts of the surrounding spectrum theory are also known. Pierce [9, Theorem 3.4], crediting Dress, identifies the rational Burnside algebra of finite continuous actions of a profinite group with locally constant rational-valued functions on a Boolean space. For $\widehat{\mathbb{Z}}$, this is the rationalization of the finite-support subring of R, whose relation to the coordinate completion is recorded in Section 2.

Parmenter–Spiegel [10] determine the prime spectra of necklace rings over fields; their results give all positive-characteristic primes of R by reduction modulo a rational prime. Levy–Loustaunau–Shapiro [11] describe the spectrum of the ambient integer product, and Finocchiaro–Frisch–Windisch [12] develop a general theory of prime ideals in products. Kionke's p-adic cardinality construction [13] supplies the fixed-point limits that occur in our closed-prime families. Mazza's treatment of profinite Burnside rings [14] makes explicit a closed-ideal convention in its discussion

of prime ideals. Our ideals throughout are ordinary algebraic ideals; closure is imposed only when stated.

The general passage between distributive lattices and spectra is standard; we use the formulation surveyed by Lombardi [15]. The principal case of (1.1) also follows from Lipman's strict-closure method [16]. The arguments developed below concern the arbitrary-ideal square bound, the realization and normalization of compatible profiles, and their consequences for this particular spectrum. The comparison with earlier constructions is therefore made at the level of these statements, without a claim that the underlying Witt–Burnside correspondence or ambient valuation theory is new.

Section 2 fixes the coordinates and proves the ghost-image criterion used throughout. Section 3 proves ideal transfer. Section 4 characterizes gcd profiles and derives the two-generator radical bound. Section 5 uses profiles to describe all primes and to compare ambient contractions. Section 6 gives the closed primes and every ambient fiber above them. A single profile supported arithmetically on a prime-power sequence will connect the uniformity issue in Section 4 with the valuation fibers in Section 6.

2. Coordinates and ghost congruences

The underlying additive group of $R = Nr(\mathbb{Z})$ is $\prod_{n\geq 1}\mathbb{Z}$. We initially denote a necklace vector by $a = (a_n)$. Addition is coordinatewise, and multiplication is

$$(ab)_n = \sum_{\operatorname{lcm}(i,j)=n} \gcd(i,j)a_i b_j. \tag{2.1}$$

Only finitely many pairs occur in each coordinate. If e_n is the vector supported at n with value one, then

$$e_i e_j = \gcd(i,j)e_{\operatorname{lcm}(i,j)}, \qquad 1_R = e_1. \tag{2.2}$$

The ghost map is

$$g : R \longrightarrow B = \prod_{n\geq 1}\mathbb{Z}, \qquad g_n(a) = \sum_{d|n} da_d. \tag{2.3}$$

It is an injective ring homomorphism. Multiplicativity follows from (2.1), since $ij = \gcd(i,j)\operatorname{lcm}(i,j)$; injectivity follows successively from the coefficient n of a_n in $g_n(a)$. The product in B is coordinatewise. Integer multiples of the unit have constant ghost sequences.

The following familiar congruence criterion is included to make the later coefficient constructions self-contained; compare [3, Example 3.4].

Lemma 2.1 (Ghost image). A sequence $x = (x_n) \in B$ belongs to $g(R)$ if and only if

$$x_n \equiv x_{n/p} \pmod{p^k} \qquad (p^k \| n). \tag{2.4}$$

Proof. For a ghost sequence, the difference $x_n - x_{n/p}$ is the sum of da_d over divisors d of n with $v_p(d) = k$. Every such term is divisible by p^k.

Conversely, suppose (2.4) holds and construct necklace coordinates in increasing order. Once a_d is known for $d < n$, put

$$a_n = \frac{1}{n}\left(x_n - \sum_{\substack{d\mid n\\ d<n}} d a_d\right). \tag{2.5}$$

For each $p^k \| n$, the sum in parentheses is congruent modulo p^k to $x_n - x_{n/p}$: the proper divisors with p-valuation below k are exactly the divisors of n/p, and all remaining terms vanish modulo p^k. Thus the numerator is divisible by every prime-power factor of n, so $a_n \in \mathbb{Z}$. This proves existence; the same recursion proves uniqueness. Equivalently, Möbius inversion gives $a_n = n^{-1}\sum_{d\mid n} \mu(n/d)x_d$. □

From Section 3 onward we identify R with $g(R) \subseteq B$. Unless explicitly called a necklace coordinate, x_n will then mean a ghost coordinate. This allows divisibility statements and coordinatewise products to be read in B. The symbols e_n continue to refer to the necklace basis vectors, whose ghosts satisfy

$$g_j(e_n) = \begin{cases} n, & n \mid j, \\ 0, & n \nmid j. \end{cases} \tag{2.6}$$

The coordinate topology τ on R is the product topology of discrete necklace coordinates. It is also the topology induced by the ghost embedding: (2.3) and the finite recursion (2.5) make both coordinate changes continuous onto the ghost image. The image is closed in B, since each congruence (2.4) is closed. In particular, (R, τ) is a complete metrizable topological ring. Finite necklace truncations converge to their underlying vector.

All ideals below consist of finite sums of products with ring elements, as in ordinary commutative algebra. We do not close them under limits. For a sequence of nonnegative integers we use $v_p(0) = \infty$, $\gcd(0, 0) = 0$, and $0 \mid t$ if and only if $t = 0$. Divisibility between sequences always means divisibility at every coordinate. The term "compatible profile" will refer to condition (1.3), not to the stronger ghost congruences (2.4).

Two related coefficient operations should also be distinguished. Reducing necklace coordinates gives $R/pR \cong Nr(\mathbb{F}_p)$, a ring of characteristic p. In Witt coordinates, the functorial coefficient map $W_{\mathrm{big}}(\mathbb{Z}) \to W_{\mathrm{big}}(\mathbb{F}_p)$ is a different map, and its p-typical component has target $\mathbb{Z}_p$. Likewise, rationalizing R does not produce the entire rational ghost product. The rational ghost vector $(1, 0, 0, ...)$ has necklace coordinates $\mu(n)/n$, whose denominators have no common bound; it is therefore not in $R \otimes \mathbb{Q}$. The infinite congruence problem persists even when finite truncations have simpler descriptions.

For the finite-set Burnside comparison, put $T = \bigoplus_{n\geq 1} \mathbb{Z}e_n \subseteq R$. The orbit multiplication (2.2) identifies T with the Burnside ring of finite continuous $\widehat{\mathbb{Z}}$-sets, and Pierce's rational algebra becomes $\mathbb{Q} \otimes_{\mathbb{Z}} T$ [9, §§1–3]. Its elements retain finite support after scalar extension. Completing T in the necklace coordinate topology gives R, whose elements may have arbitrary integer coordinate sequences. The arguments below concern ordinary ideals in this integral completion.

2.1. The known positive-characteristic fibers

For orientation, the primes containing a rational prime already have a direct description. Fix p and put $S_p = \{m \geq 1 : p \nmid m\}$. The results of Parmenter–Spiegel [10, Propositions 2.4 and

4.2–4.4, and p. 1631] identify the reduced quotient of $Nr(\mathbb{F}_p)$ with $\prod_{m\in S_p}\mathbb{F}_p$: the ghost kernel is square-zero, and the ghost image is this product. It follows that

$$V(pR) \cong \operatorname{Spec}\left(\prod_{m\in S_p}\mathbb{F}_p\right). \tag{2.7}$$

Its primes, all maximal, are indexed by ultrafilters $\mathcal{U}$ on S_p:

$$\mathfrak{m}_{p,\mathcal{U}} = \{a \in R : \{m \in S_p : g_m(a) \equiv 0 \pmod p\} \in \mathcal{U}\}. \tag{2.8}$$

Here an ultrafilter is a family of subsets closed under finite intersections and supersets that contains exactly one of each subset and its complement. It is principal if its members are the sets containing a fixed index. A nonprincipal ultrafilter on a countable set contains every cofinite subset. Each residue field in (2.8) is $\mathbb{F}_p$, since a sequence taking values in this finite field takes one value on an ultrafilter-large set.

Principal ultrafilters give the open primes $M_{m,p} = \ker(g_m \bmod p)$ used later. Nonprincipal ultrafilters give dense ideals in the coordinate topology. Indeed, their kernels in the product contain every finitely supported function. The map $R \to \prod_{m\in S_p}\mathbb{F}_p$ is continuous, surjective and open: reduction on the p-free necklace coordinates is followed by an invertible triangular divisor-sum transformation. The inverse image of each of these dense kernels is therefore dense. Thus the known positive-characteristic description already supplies primes outside the closed families. The subsequent results include these primes and also the characteristic-zero part of the ordinary spectrum.

3. Ideal transfer

Throughout this section we identify the necklace ring with its ghost image $R \subseteq B = \prod_{n\geq 1}\mathbb{Z}$. Thus a_n denotes the n-th ghost coordinate of $a \in R$, and multiplication is coordinatewise. The criterion from Section 2 takes the form

$$a_n \equiv a_{n/p} \pmod{p^{v_p(n)}} \qquad (p \mid n). \tag{3.1}$$

For $a_1, \dots, a_r \in R$, write $a_{i,n}$ for the n-th coordinate of a_i, and put

$$d_n = \gcd(a_{1,n}, \dots, a_{r,n}) \geq 0.$$

An element $b \in B$ belongs to $(a_1, \dots, a_r)B$ exactly when $d_n \mid b_n$ for every n, with the convention $0 \mid b_n$ if and only if $b_n = 0$. Indeed, the forward implication follows coordinatewise. Conversely, Bézout's identity supplies coefficients at each coordinate; collecting the coefficient of a_i over all coordinates gives one element of B, so there are still just r summands in the resulting expression for b.

The same choice of coefficients need not lie in R, since unrelated coordinatewise choices need not satisfy (3.1). We shall construct coefficients satisfying those congruences when the target is a product bc, where both factors belong to the contracted extended ideal. The additional divisibility supplied by the two factors permits each coordinate equation to be corrected without disturbing the previously imposed congruences.

Theorem 3.1 (Ideal transfer). For every ideal $I \subseteq R$,

$$(IB \cap R)^2 \subseteq I. \tag{3.2}$$

More explicitly, if $r \geq 1$, $a_1, \dots, a_r, b, c \in R$, and

$$d_n \mid b_n, \qquad d_n \mid c_n \qquad (n \geq 1), \tag{3.3}$$

then there are $h_1, \dots, h_r \in R$ such that

$$bc = \sum_{i=1}^{r} a_i h_i. \tag{3.4}$$

Proof. We first prove the explicit assertion. Construct integers $h_{i,n}$ in increasing order of n, requiring

$$\sum_{i=1}^{r} a_{i,n} h_{i,n} = b_n c_n, \tag{3.5}$$

and, in addition, $d_n \mid h_{i,n}$ when $d_n > 0$, while $h_{i,n} = 0$ when $d_n = 0$. At each stage we also require

$$h_{i,n} \equiv h_{i,n/p} \pmod{p^k} \qquad (p^k \| n). \tag{3.6}$$

The divisibility requirement on $h_{i,n}$ is an auxiliary condition for the induction. It will ensure that the congruences at the next stage are compatible and that their correction has enough divisibility.

Suppose that all coordinates below n have been constructed. If $d_n = 0$, all $a_{i,n}$ vanish, and (3.3) gives $b_n = c_n = 0$. Set $h_{i,n} = 0$. For $p^k \| n$, (3.1) gives $p^k \mid a_{i,n/p}$ for every i. If $d_{n/p} = 0$, the preceding coefficients are zero by construction. Otherwise $p^k \mid d_{n/p} \mid h_{i,n/p}$. Thus (3.6) holds in both cases, and (3.5) holds because both sides vanish.

Now suppose $d = d_n > 0$, and set $L = \operatorname{lcm}(n, d)$. First choose preliminary coefficients $h_i^0 \in \mathbb{Z}$ satisfying

$$h_i^0 \equiv 0 \pmod{d}, \qquad h_i^0 \equiv h_{i,n/p} \pmod{p^k} \quad (p^k \| n). \tag{3.7}$$

To justify this choice, fix $p^k \| n$, and write $\alpha = v_p(d)$. When $\alpha < k$, the congruences $a_{i,n/p} \equiv a_{i,n} \pmod{p^k}$ imply

$$v_p(d_{n/p}) = \alpha. \tag{3.8}$$

Every $a_{i,n/p}$ is divisible by p^α, and at least one has valuation exactly α, because the same holds at n and adding a multiple of p^k cannot change that valuation. In particular, $d_{n/p} > 0$. If $\alpha \geq k$, instead every $a_{i,n/p}$ is divisible by p^k; here $d_{n/p}$ may be zero. The induction requirements therefore give, in either case,

$$p^{\min(\alpha,k)} \mid h_{i,n/p}.$$

This is precisely the compatibility condition between the congruence modulo d and the congruence modulo p^k in (3.7). Congruences associated with different primes are independent. The Chinese remainder theorem supplies h_i^0, with all choices differing by multiples of L.

These preliminary coefficients satisfy the required congruences but need not yet satisfy (3.5). Let their residual be

$$E = b_n c_n - \sum_{i=1}^{r} a_{i,n} h_i^0.$$

We claim that $dL \mid E$. Fix a prime p, and write $\alpha = v_p(d)$, $k = v_p(n)$, allowing $k = 0$. There are two cases.

If $\alpha \geq k$, each of $a_{i,n}, h_i^0, b_n, c_n$ is divisible by p^α, using (3.3) and (3.7). Consequently $p^{2\alpha} \mid E$. This is the required power, since

$$v_p(dL) = \alpha + \max(\alpha, k) = 2\alpha.$$

If $\alpha < k$, put $m = n/p$. The equation already established at m gives

$$\begin{aligned} E = (b_n c_n - b_m c_m) - \sum_{i=1}^{r} (a_{i,n} - a_{i,m}) h_{i,m} \\ - \sum_{i=1}^{r} a_{i,n} (h_i^0 - h_{i,m}). \end{aligned} \tag{3.9}$$

Every term on the right is divisible by $p^{k+\alpha}$. For the first term, use

$$b_n c_n - b_m c_m = (b_n - b_m) c_n + b_m (c_n - c_m).$$

Both differences are divisible by p^k by (3.1), while c_n and b_m are divisible by p^α, the latter by (3.3) and (3.8). In the second sum of (3.9), each coordinate difference is divisible by p^k, and $p^\alpha \mid h_{i,m}$ by induction and (3.8). In the last sum, $p^\alpha \mid a_{i,n}$, and $p^k \mid h_i^0 - h_{i,m}$ by (3.7). Thus $p^{k+\alpha} \mid E$, which again is the required power $p^{v_p(dL)}$.

This proves $dL \mid E$, including primes dividing d but not n, which occur in the first case. Since the integers $a_{i,n}/d$ have gcd one, Bézout's identity gives integers $z_1, \ldots, z_r$ with

$$\sum_{i=1}^{r} \frac{a_{i,n}}{d} z_i = \frac{E}{dL}.$$

Set $h_{i,n} = h_i^0 + L z_i$. This changes the sum in (3.5) by E, preserves (3.7), and preserves divisibility by d. The induction is complete. It includes $n = 1$: there are no predecessor congruences, $L = d$, and $d^2 \mid b_1 c_1$.

For each i, (3.6) and the ghost-image criterion give an element $h_i = (h_{i,n})_{n \geq 1} \in R$. Equation (3.5) then gives (3.4). Although the construction has infinitely many coordinate stages, its conclusion has exactly r coefficients and r summands. No closure assumption on an ideal, and no infinite sum of ideal elements, enters this conclusion.

Finally, take $b, c \in IB \cap R$. Their two expressions in IB involve only finitely many elements of I. Combine these into one list $a_1, \ldots, a_r$; then (3.3) holds, so (3.4) gives $bc \in I$. The zero ideal case is immediate, and finite sums of such products give (3.2). □

The use of two possibly different targets proves the ideal-square containment directly. Knowing only that each element of $IB \cap R$ has its square in I would not justify (3.2) by polarization over this ring.

Corollary 3.2 (Radical contraction and prime lifting). For every ideal $I \subseteq R$,

$$\sqrt{IB} \cap R = \sqrt{I}. \tag{3.10}$$

Every prime ideal of R is the contraction of a prime ideal of B.

Proof. If $b \in \sqrt{IB} \cap R$, some b^m belongs to $IB \cap R$. Theorem 3.1 gives $b^{2m} \in I$, and hence $b \in \sqrt{I}$. The reverse inclusion follows from $I \subseteq IB$.

For a prime $P \subseteq R$, Theorem 3.1 gives $PB \cap R = P$: an element of the left-hand side has its square in P, and therefore belongs to P. Set $S = R \setminus P$. Since $PB \cap S = \emptyset$, the ideal $S^{-1}PB$ is proper in $S^{-1}B$. Choose a maximal ideal containing it and contract this ideal to B. The resulting prime Q contains PB and avoids S, so $Q \cap R = P$. □

For finitely generated ideals, radical contraction translates membership into an integer divisibility condition. A single power must satisfy this condition at every coordinate.

Corollary 3.3 (Uniform radical criterion). Let $I = (a_1, \dots, a_r)R$, with $r \geq 1$, and let $d_n = \gcd_i a_{i,n}$. Then, for every $b \in R$,

$$b \in \sqrt{I} \quad \Longleftrightarrow \quad \exists N \geq 1 \ \forall n \geq 1, \qquad d_n \mid b_n^N. \tag{3.11}$$

Proof. If $b^N \in I$, taking coordinates proves the divisibilities. Conversely, the divisibilities imply $b^N \in IB \cap R$, so Theorem 3.1 gives $b^{2N} \in I$. □

The same exponent N must work at every coordinate. A coordinate-dependent choice of exponent does not imply (3.11). Section 4 will characterize which sequences (d_n) occur and give an example showing why this uniformity cannot be dropped.

Remark 3.4 (The exponent and the principal case). The containment $IB \cap R \subseteq I$ fails even for a principal ideal. Let e_2 be the necklace basis element whose second necklace coordinate is one and whose other necklace coordinates are zero. Its ghost coordinates are two at even indices and zero at odd indices, so $e_2 \in 2B \cap R$. It does not belong to $2R$, since multiplication by the integer two doubles every necklace coordinate. Coordinatewise multiplication gives $e_2^2 = 2e_2 \in 2R$. Thus the exponent two in Theorem 3.1 cannot be replaced uniformly by one.

The principal-ideal case also follows from the established strict-closure method in [16, §4, proof of Proposition 4.5]. Recall that an extension $A \subseteq C$ is strictly closed if

$$A = \{x \in C : x \otimes 1 = 1 \otimes x \text{ in } C \otimes_A C\}.$$

Here R is strictly closed in B: for each $p^k \| n$, evaluation at n and at n/p, followed by reduction modulo p^k, agrees on R. The resulting map $B \otimes_R B \to \mathbb{Z}/p^k\mathbb{Z}$ sends $x \otimes 1$ to x_n and $1 \otimes x$ to $x_{n/p}$. Tensor equality therefore implies (3.1) and hence membership in R. If $b = ax$ and $c = ay$ lie in $aB \cap R$, put $h = axy$. Then

$$h \otimes 1 = x \otimes ay = ax \otimes y = 1 \otimes h,$$

using $ay = c \in R$ and $ax = b \in R$. Consequently $h \in R$ and $bc = ah \in aR$. This applies Lipman's tensor argument; the arbitrary-ideal assertion above is proved by the compatible coefficient construction.

Proposition 3.5. The extension $R \subseteq B$ is not integral.

Proof. Let $x_n = \Omega(n)$, where $\Omega(n)$ counts prime factors with multiplicity and $\Omega(1) = 0$. Suppose that $x \in B$ satisfies a monic equation of degree $D \geq 1$ with coefficients $a_0, \dots, a_{D-1} \in R$. For every prime p, repeated application of (3.1) gives

$$a_{j,p^k} \equiv a_{j,1} \pmod{p} \qquad (k \geq 1).$$

Since $x_{p^k} = k$, reduction of these coordinate equations modulo p says that the single monic polynomial

$$T^D + \sum_{j=0}^{D-1} \overline{a_{j,1}} T^j \in \mathbb{F}_p[T]$$

vanishes at every residue class in $\mathbb{F}_p$. Taking $p > D$ contradicts the bound on the number of roots of a nonzero polynomial over a field. □

Thus prime lifting here holds for an extension that is not integral. Theorem 3.1 supplies the needed passage from coordinatewise divisibility to ideal membership. The next question is which gcd sequences can serve as the numerical data in Corollary 3.3; answering it will reduce finitely generated radicals to two generators and give an arithmetic description of the quasicompact Zariski opens.

4. Compatible profiles and the Zariski lattice

For a finitely generated ideal $I = (a_1, \dots, a_r)$, the radical criterion of Section 3 depends on the generators through the nonnegative integers

$$d_I(n) = \gcd(g_n(a_1), \dots, g_n(a_r)).$$

The profile d_I is independent of the generating list, since $IB = d_I B$. We now determine which numerical sequences occur in this way. This removes the need to specify ring elements before applying the radical criterion and, at the same time, bounds the number of elements needed to define their common zero set.

Throughout, gcds are nonnegative, $\gcd(0, \dots, 0) = 0$, and the empty gcd is 0. We set $v_p(0) = \infty$; divisibility of profiles is coordinatewise, with $0 \mid x$ meaning $x = 0$.

Definition 4.1. A profile $d = (d_n)_{n \geq 1} \in \mathbb{N}_0^{\mathbb{N}_{>0}}$ is **compatible** if

$$\min\{v_p(d_n), k\} = \min\{v_p(d_{n/p}), k\} \qquad \text{whenever } p^k \| n. \tag{4.1}$$

Write $\mathcal{D}$ for the set of compatible profiles.

The truncation in (4.1) records what survives on taking gcds in the ghost congruence $g_n(a) \equiv g_{n/p}(a) \pmod{p^k}$. Valuations below k must agree; valuations at least k need not agree. No congruence between the positive integers d_n and $d_{n/p}$ themselves is required.

Theorem 4.2 (Profile realization). A nonnegative integer profile belongs to $\mathcal{D}$ if and only if it is the coordinatewise gcd of finitely many ghost sequences. Every profile in $\mathcal{D}$ is the gcd of two ghost sequences.

Proof. Suppose first that $d_n = \gcd_i g_n(a_i)$. For $p^k \| n$, each pair $g_n(a_i), g_{n/p}(a_i)$ agrees modulo p^k, so its valuations have the same truncation at k. Taking the minimum over i gives (4.1).

Conversely, let d satisfy (4.1). We construct two sequences $A_{1,n}, A_{2,n} \in \mathbb{Z}$, by induction on n, such that

$$\gcd(A_{1,n}, A_{2,n}) = d_n, \qquad A_{i,n} \equiv A_{i,n/p} \pmod{p^k} \quad (p^k \| n). \tag{4.2}$$

All predecessor values have been chosen when we reach n. If $d_n = 0$, set $A_{1,n} = A_{2,n} = 0$. Equation (4.1) says that the predecessor gcd is divisible by p^k, so both predecessor entries are divisible by p^k, as required.

Suppose $d = d_n > 0$. For each i, impose divisibility by d together with the congruences in (4.2). These conditions are consistent: if $p^k \| n$, their common modulus has p-valuation $\min(v_p(d), k)$, and (4.1) says that this power of p divides each predecessor entry. The Chinese remainder theorem therefore specifies one residue class for $A_{i,n}$ modulo $L = \operatorname{lcm}(n, d)$. After writing $A_{1,n} = du$ and $A_{2,n} = dv$, we obtain

$$u \equiv u_0 \pmod{M}, \qquad v \equiv v_0 \pmod{M}, \qquad M = L/d = n/\gcd(n, d). \tag{4.3}$$

It remains to choose representatives with $\gcd(u, v) = 1$.

The prescribed pair is primitive modulo M. Indeed, if $p \mid M$, put $\alpha = v_p(d) < k = v_p(n)$. Compatibility gives predecessor gcd valuation exactly α, so at least one predecessor entry has valuation α. Dividing its congruence by p^α, and then by the unit d/p^α, shows that at least one of u_0, v_0 is nonzero modulo p. Thus $\gcd(u_0, v_0, M) = 1$.

To lift this pair, choose a nonzero $u \equiv u_0 \pmod{M}$. For each prime $\ell \mid u$ with $\ell \nmid M$, choose a residue of t modulo ℓ for which $v_0 + Mt \not\equiv 0 \pmod{\ell}$. There are finitely many such primes, and the Chinese remainder theorem supplies one t meeting all these requirements. If instead $\ell \mid u$ and $\ell \mid M$, primitivity already gives $\ell \nmid v_0 + Mt$. Hence $v = v_0 + Mt$ is coprime to u. When $M = 1$, one may simply take $u = 1$.

The values du, dv complete the induction. Both resulting sequences satisfy the ghost congruences, so the ghost-image criterion of Section 2 identifies them with ghost sequences of elements of R. Their gcd profile is d. □

The realization is global: the same two elements work at every coordinate. The recursive choice avoids introducing a common prime factor at the current coordinate while preserving every congruence with the earlier coordinates. The theorem does not assert that d itself is a ghost sequence.

The Chinese remainder step for choosing each new entry has a counterpart for finite graphs in Dilaver and Altınok [17, proof of Theorem 3.3], where integer values at vertices satisfy edge congruences and prescribed divisibility conditions. The construction above also controls the gcd of the two entries at each new coordinate while extending the same two sequences throughout.

For example, fix a prime p and consider

$$h_{p^k} = p^{k+1} \quad (k \geq 0), \qquad h_n = 1 \quad \text{when } n \text{ is not a power of } p. \tag{4.4}$$

On the p-power chain, the two valuations at an edge both reach its truncation threshold; on every other relevant prime chain, both truncated valuations are zero. Thus h is compatible. For $p = 2$, however, $h_1 = 2$ and $h_3 = 1$ fail the ghost congruence modulo 3. Theorem 4.2 realizes this numerical sequence as the gcd of two ghosts even though it is not itself a ghost.

We will use two closure properties of $\mathcal{D}$. Coordinatewise gcd and finite multiplication preserve compatibility, because truncated valuations commute with minima and satisfy

$$\min(x + y, k) = \min(\min(x, k) + \min(y, k), k).$$

Arbitrary coordinatewise gcds also preserve compatibility: taking an infimum of valuations commutes with truncation at a finite integer. These statements include zero entries and the empty gcd convention.

4.1. Radicals and two generators

For each $d \in \mathcal{D}$, choose a two-generated ideal J_d with profile d. The radical criterion from Section 3 reads

$$a \in \sqrt{J_d} \quad \Longleftrightarrow \quad \exists N \geq 1 \ \forall n \geq 1, \qquad d_n \mid g_n(a)^N. \tag{4.5}$$

In particular, $\sqrt{J_d}$ does not depend on the chosen realization.

The exponent in (4.5) cannot vary with n. For the profile h in (4.4), the element $p\,1_R$ has constant ghost value p. At each coordinate some power of p is divisible by h_n, but at $n = p^k$ the necessary exponent is at least $k+1$. Consequently $p\,1_R \notin \sqrt{J_h}$. Coordinatewise prime supports alone therefore do not determine radicals. The growth of the required exponents across infinitely many coordinates remains visible in the spectrum.

Corollary 4.3. Every finitely generated ideal $I \subseteq R$ has arithmetical rank at most two. More precisely, there are $u, v \in I$ with

$$\sqrt{I} = \sqrt{(u, v)}. \tag{4.6}$$

Proof. Realize d_I by $h_1, h_2 \in R$. Since $h_i \in d_I B \cap R = IB \cap R$, the square theorem of Section 3 gives $h_i^2 \in I$. The ideal (h_1^2, h_2^2) has profile d_I^2, and (4.5) gives the same radical for profiles d_I and d_I^2. Take $u = h_1^2$ and $v = h_2^2$. □

Thus every closed set defined by finitely many elements is defined by two elements, and every quasicompact Zariski open is a union of two principal opens. The assertion concerns finitely generated ideals and uses no finite-dimensionality hypothesis.

4.2. The arithmetic lattice

For $d, e \in \mathcal{D}$, define

$$d \sim e \quad \Longleftrightarrow \quad \exists N \geq 1 \ \forall n, \qquad e_n \mid d_n^N \text{ and } d_n \mid e_n^N. \tag{4.7}$$

By (4.5), this is equivalent to $\sqrt{J_d} = \sqrt{J_e}$. To see that one common exponent suffices, apply (4.5) to the finitely many generators of each ideal, and take the maximum of their exponents. Conversely, divisibility of the gcd of the N-th powers is equivalent to divisibility of each such power. This argument also proves the corresponding one-sided comparison.

Proposition 4.4. The assignment $[d] \mapsto D(J_d)$ identifies $\mathcal{D}/\!\sim$ with the lattice of quasicompact Zariski opens of $\operatorname{Spec} R$. Its order and operations are

$$\begin{aligned} &[d] \leq [e] \Longleftrightarrow \exists N \geq 1 \ \forall n, \quad e_n \mid d_n^N, \\ &[d] \vee [e] = [\gcd(d, e)], \\ &[d] \wedge [e] = [de]. \end{aligned} \tag{4.8}$$

The constant profiles 0 and 1 are respectively the bottom and top elements.

Proof. Every quasicompact open is $D(I)$ for a finitely generated ideal I, and Theorem 4.2 identifies its profile with a member of $\mathcal{D}$. The equivalence and order follow from (4.5), since $D(I) \subseteq D(J)$ precisely when $I \subseteq \sqrt{J}$. The profiles of $I + J$ and IJ are respectively the coordinatewise gcd and product of their profiles. Now use $D(I+J) = D(I) \cup D(J)$ and $D(IJ) = D(I) \cap D(J)$. □

This is an arithmetic presentation of the usual Zariski lattice [15]. Its ring-specific content is the condition (4.1), together with the uniform divisibility relation (4.8). We next use that arithmetic both to compare contractions from B and to describe individual points.

5. Normalization, contractions, and points

An arbitrary generator $d \in \mathbb{N}_0^{\mathbb{N}_{>0}}$ of a principal ideal of B need not be compatible. Nevertheless, any ghost sequence divisible by d has a compatible absolute-value profile. The contraction $dB \cap R$ therefore depends on the compatible divisibility conditions forced by d. These conditions have a least representative.

5.1. Least compatible multiples

Define $\sigma(d)$ to be the least compatible multiple of d, with respect to coordinatewise divisibility. It exists because the all-zero profile is a compatible multiple and the coordinatewise gcd of all compatible multiples is compatible. Thus

$$d \mid \sigma(d), \qquad \sigma(d) \mid e \quad \text{for every compatible } e \text{ with } d \mid e. \tag{5.1}$$

The operator σ is monotone and idempotent.

Proposition 5.1 (Normalization). For every nonnegative profile d,

$$dB \cap R = \sigma(d)B \cap R, \qquad \sqrt{dB} \cap R = \sqrt{J_{\sigma(d)}}. \tag{5.2}$$

Moreover, for every integer $N \geq 1$,

$$\sigma(d) \mid \sigma(d^N) \mid \sigma(d)^N. \tag{5.3}$$

Proof. If $a \in dB \cap R$, its absolute ghost profile $(|g_n(a)|)_n$ is a compatible multiple of d, hence is divisible by $\sigma(d)$. This proves the first equality. For the second, apply the same argument to the compatible profile $(|g_n(a)|^N)_n$ and then use (4.5). The reverse implications follow from $d \mid \sigma(d)$. Finally, monotonicity gives the first divisibility in (5.3), while $\sigma(d)^N$ is a compatible multiple of d^N, giving the second. □

In particular, σ preserves uniform power equivalence: if arbitrary ambient profiles d, e satisfy (4.7), monotonicity and (5.3) give $\sigma(d) \sim \sigma(e)$.

5.2. Equality and containment of contractions

For an ideal $Q \subseteq B$, write

$$\mathcal{D}_Q = Q \cap \mathbb{N}_0^{\mathbb{N}_{>0}}, \qquad P = Q \cap R.$$

The set $\mathcal{D}_Q$ consists of arbitrary nonnegative ambient profiles; its elements need not satisfy (4.1). If Q is given by ultrafilter or valuation data, these data determine membership in $\mathcal{D}_Q$ directly.

Theorem 5.2 (Comparison of contractions). For every ideal $Q \subseteq B$,

$$PB = \bigcup_{d \in \mathcal{D}_Q} \sigma(d)B. \tag{5.4}$$

For ideals $Q, Q' \subseteq B$, with contractions P, P',

$$P \subseteq P' \quad \Longleftrightarrow \quad \forall d \in \mathcal{D}_Q \; \exists e \in \mathcal{D}_{Q'}, \qquad \sigma(e) \mid \sigma(d). \tag{5.5}$$

Consequently, equality of contractions is equivalent to the two normalized profile families being cofinal under divisibility in both directions.

Proof. Given $d \in \mathcal{D}_Q$, realize $\sigma(d)$ by ghosts of $u, v \in R$. Both are divisible coordinatewise by d, so $u, v \in Q \cap R = P$. Their gcd generates $(u, v)B = \sigma(d)B$, proving one inclusion in (5.4).

For the reverse inclusion, take $a \in P$. Multiplication by a coordinatewise sign sequence shows that $d(a) = (|g_n(a)|)_n$ belongs to Q. It is compatible, so $\sigma(d(a)) = d(a)$, and a belongs to the

displayed union. That union is an ideal: if $d, e \in \mathcal{D}_Q$, coordinatewise Bézout coefficients show that $\gcd(d, e) \in \mathcal{D}_Q$, and monotonicity gives

$$\sigma(\gcd(d, e)) \mid \sigma(d), \sigma(e).$$

Thus the principal ideals in the union form a directed family.

The right side of (5.5) is precisely inclusion of the unions in (5.4): for the forward implication of this assertion, test the generator $\sigma(d)$ of each principal ideal. Finally, for any contracted ideal $P = Q \cap R$,

$$P \subseteq PB \cap R \subseteq Q \cap R = P.$$

Extension and contraction therefore identify $PB \subseteq P'B$ with $P \subseteq P'$, proving (5.5). □

No power exponent occurs in (5.5), because it compares the extended ideals themselves. This differs from the radical comparison (4.8). The theorem applies to all ambient ideals; the square theorem is needed separately to ensure that every prime of R is a contraction of an ambient prime.

5.3. Numerical propagation

The normalizer is numerical: its definition quantifies over profiles satisfying (4.1), rather than over ideals of R. The following formula describes the propagation of those conditions, including the possibility that infinitely many required prime factors force an entry to vanish.

Let Z be the least set of positive integers containing $\{n : d_n = 0\}$ and closed under the rule

$$\left.\begin{array}{c}\text{for infinitely many primes } p \nmid n, \text{ there is } j \geq 1 \\ \text{with } p \mid d_{np^j} \text{ or } np^j \in Z\end{array}\right\} \quad \Longrightarrow \quad n \in Z. \tag{5.6}$$

Put $\alpha(n, p) = \infty$ for $n \in Z$, and $\alpha(n, p) = v_p(d_n)$ otherwise. For $n = mp^k$, $p \nmid m$, set

$$\begin{aligned} c(k, j) &= \begin{cases} \infty, & k = j, \\ \min(k, j) + 1, & k \neq j, \end{cases} \\ \beta(n, p) &= \sup_{j \geq 0} \min\{\alpha(mp^j, p), c(k, j)\}. \end{aligned} \tag{5.7}$$

Then

$$\sigma(d)_n = \begin{cases} 0, & n \in Z, \\ \displaystyle\prod_p p^{\beta(n,p)}, & n \notin Z. \end{cases} \tag{5.8}$$

The product in the second case has finite support and finite exponents. Appendix A proves the formula and constructs Z by transfinite iteration. The cap $c(k, j)$ is the smallest congruence threshold along the path from level j to level k on one p-chain. Equation (5.6) accounts for a different requirement: an integer divisible by infinitely many primes must be zero.

Three examples show how these two operations affect contraction. If $d_m = p$ at one index and $d_n = 1$ elsewhere, then $\sigma(d)_n = p$ precisely when n and m have the same p-free part. Indeed, valuation one propagates along that entire p-chain, and no other valuation is introduced. Hence the contractions of the characteristic-p coordinate primes agree along the chain.

If instead $d_m = 0$ and all other entries are 1, no additional zero is forced. For $n \neq m$, the normalized value is 1 unless n/m or m/n is a power of one prime p; in that case it is

$$\sigma(d)_n = p^{\min(v_p(n), v_p(m))+1}. \tag{5.9}$$

These are the valuations propagated from a zero along each prime chain through m. The contraction is $\ker g_m$.

Finally, if $d_n = n$, each n has infinitely many branches nq with $q \nmid n$ and $q \mid d_{nq}$. Thus every index enters Z, and $\sigma(d) = 0$. It follows that $dB \cap R = 0$, and also $\sqrt{dB} \cap R = 0$, since R is reduced. In this example normalization changes a profile with no zero entries into the zero profile.

5.4. Points from finite arithmetic conditions

The comparison theorem identifies when different ambient descriptions have the same contraction. We can also describe a point without making an ambient choice. A prime determines which of the finitely generated radicals represented by profiles it contains. We record these containment decisions by a function

$$\varepsilon_P(d) = \begin{cases} 1, & J_d \subseteq P, \\ 0, & J_d \not\subseteq P, \end{cases} \qquad d \in \mathcal{D}. \tag{5.10}$$

The function is independent of the chosen J_d, and

$$\varepsilon_P(d(a)) = 1 \quad \Longleftrightarrow \quad a \in P, \qquad d(a) = (|g_n(a)|)_n. \tag{5.11}$$

We call ε_P the profile signature of P. A value of one records containment in the prime, rather than membership in a Zariski open.

Proposition 5.3 (Finite patterns). Let $A, C \subseteq \mathcal{D}$ be finite, and put

$$a_n = \gcd_{d \in A} d_n, \qquad c_n = \prod_{e \in C} e_n, \tag{5.12}$$

with empty gcd 0 and empty product 1. A prime signature takes the value one on A and zero on C if and only if

$$\forall N \geq 1 \ \exists n \geq 1, \qquad a_n \nmid c_n^N. \tag{5.13}$$

Proof. Set $I_A = \sum_{d \in A} J_d$ and $K_C = \prod_{e \in C} J_e$, with $I_\emptyset = 0$ and $K_\emptyset = R$. Their profiles are a, c. For a prime P, requiring every J_e, $e \in C$, to avoid containment in P is equivalent to $K_C \not\subseteq P$. Prime ideal separation therefore gives the required prime exactly when $K_C \not\subseteq \sqrt{I_A}$: choose an element of $K_C \setminus \sqrt{I_A}$ and separate it from I_A.

Choose a finite generating list $b_1, \dots, b_t$ for K_C, using (0) or (1) in the two degenerate cases. By (4.5), containment in $\sqrt{I_A}$ is equivalent to the existence of a common $N \geq 1$ such that $a_n \mid g_n(b_i)^N$ for every n, i. The exponent can be taken common because there are finitely many generators. Since

$$\gcd_i (g_n(b_i)^N) = c_n^N,$$

this condition is $\exists N \, \forall n, \ a_n \mid c_n^N$. Its negation is (5.13). □

Define X to be the set of functions $\varepsilon : \mathcal{D} \to \{0, 1\}$ for which (5.13) holds whenever $A \subseteq \varepsilon^{-1}(1)$ and $C \subseteq \varepsilon^{-1}(0)$ are finite. This definition uses only integer gcds, products, and divisibility.

Theorem 5.4 (Arithmetic points). The map $P \mapsto \varepsilon_P$ is a bijection from $\operatorname{Spec} R$ to X. Its inverse is

$$P_\varepsilon = \{a \in R : \varepsilon(d(a)) = 1\}. \tag{5.14}$$

Proof. Proposition 5.3 puts every prime signature in X, while (5.11) proves injectivity. Conversely, every finite set of values of $\varepsilon \in X$ is realized by a prime. The patterns involving the constant profiles force $\varepsilon(0) = 1$ and $\varepsilon(1) = 0$, so P_ε contains zero and excludes one. To check addition, realize simultaneously the prescribed values on $d(a), d(b), d(a+b)$; the realizing prime shows that $a, b \in P_\varepsilon$ imply $a + b \in P_\varepsilon$. A realizing prime for $d(a), d(ra)$ proves closure under

multiplication by $r \in R$, including additive inverses when $r = -1$. Likewise, the three profiles $d(a), d(b), d(ab)$ prove primality.

It remains to recover the values on arbitrary profiles. Write $J_d = (u, v)$, and realize the finite values on $d, d(u), d(v)$. In the realizing prime, J_d is contained precisely when both u and v belong. Hence

$$\varepsilon(d) = 1 \quad \Longleftrightarrow \quad \varepsilon(d(u)) = \varepsilon(d(v)) = 1,$$

which is equivalent to $J_d \subseteq P_\varepsilon$. Thus $\varepsilon_{P_\varepsilon} = \varepsilon$. □

The reconstruction is the usual passage from the Zariski lattice to its prime points [15]; Proposition 5.3 supplies its arithmetic finite-pattern test in the present ring. Each point has one signature, without a further quotient of the parameter space.

5.5. Topology and specialization

Under Theorem 5.4, the quasicompact Zariski opens are

$$U_d = \{\varepsilon \in X : \varepsilon(d) = 0\}, \qquad d \in \mathcal{D}. \tag{5.15}$$

They form a basis, and Proposition 4.4 gives $U_d \cup U_e = U_{\gcd(d,e)}$ and $U_d \cap U_e = U_{de}$. Equivalently,

$$\varepsilon(\gcd(d, e)) = \min(\varepsilon(d), \varepsilon(e)), \qquad \varepsilon(de) = \max(\varepsilon(d), \varepsilon(e)). \tag{5.16}$$

Signatures agree on uniformly power-equivalent profiles by (4.7).

The patch topology, generated by quasicompact Zariski opens and their complements, is therefore the subspace topology from the product $\{0, 1\}^{\mathcal{D}}$. A failure of (5.13) is witnessed by a finite pattern, so the complement of X is a union of open cylinders. Thus X is closed in that product and is compact Hausdorff. More explicitly, if X_F is the finite set of patterns on a finite $F \subseteq \mathcal{D}$ satisfying (5.13), then

$$X = \varprojlim_{F \subseteq \mathcal{D} \text{ finite}} X_F. \tag{5.17}$$

The restriction maps are surjective, since Proposition 5.3 realizes each permitted finite pattern by a prime and thus extends it to any larger finite set.

Specialization is also numerical:

$$P_\varepsilon \subseteq P_\eta \quad \Longleftrightarrow \quad \varepsilon(d) \le \eta(d) \quad \text{for every } d \in \mathcal{D}. \tag{5.18}$$

One implication follows from ideal containment; the other follows by testing $d(a)$ for each $a \in P_\varepsilon$. In particular, η is a specialization of ε exactly under these coordinatewise inequalities.

Appendix B translates these signatures into the ultrafilter and valuation parameters used for ambient product spectra. That alternative representation is not needed for Theorem 5.4.

The formulas are infinitary arithmetic criteria. They retain the uniform growth information already seen in (4.4), which no fixed finite collection of coordinates can record. In Section 6, the same profile h will occur as the generator of $(\ker L_{1,p})B$; its growth along the p-power chain will then determine which ambient valuation primes lie above that closed prime.

6. Closed primes and all their ambient lifts

The arithmetic description of the preceding sections applies to the ordinary prime spectrum of R. We now identify the primes that are closed for the necklace coordinate topology and compute every prime of B contracting to each of them. Closedness in this section always refers to the product topology on the necklace coordinates, with each copy of $\mathbb{Z}$ discrete. It is a condition on an ideal as a subset of R, separate from the Zariski topology on $\operatorname{Spec} R$.

The classification begins with the quotient ring. If a prime is closed, the images of the necklace basis vectors make its quotient a completion of the integer subring. The domain condition then leaves three possible quotients. After identifying their kernels, we calculate their extended ideals in B; those calculations determine the coordinate sets and valuation growth that can occur in a fiber.

6.1. The closed prime ideals

Recall that e_n denotes the necklace vector supported at n, so that $e_1 = 1$ and

$$e_i e_j = \gcd(i,j) e_{\mathrm{lcm}(i,j)}, \qquad e_n^2 = n e_n. \tag{6.1}$$

For a rational prime p and an integer m prime to p, the ghost congruences show that $g_{mp^k}(a)$ is a Cauchy sequence in $\mathbb{Z}_p$. Its limit is the following continuous ring homomorphism, where the series is written in the necklace coordinates a_n:

$$L_{m,p}(a) = \lim_{k\to\infty} g_{mp^k}(a) = \sum_{d\mid m} \sum_{k\geq 0} d p^k a_{dp^k}. \tag{6.2}$$

The last expression also shows continuity directly: modulo p^r, it depends on only finitely many necklace coordinates. Multiplicativity follows by taking the p-adic limit of the ghost homomorphisms. We use the notation

$$A_m = \ker g_m, \qquad M_{m,p} = \ker(g_m \bmod p), \qquad B_{m,p} = \ker L_{m,p}. \tag{6.3}$$

Theorem 6.1 (Closed primes). Every coordinate-topology-closed prime of R occurs exactly once in the following table.

Prime ideal	Parameters	Quotient ring	Coordinate topology
A_m	$m \geq 1$	$\mathbb{Z}$	Open
$M_{m,p}$	p prime, $p \nmid m$	$\mathbb{F}_p$	Open
$B_{m,p}$	p prime, $p \nmid m$	$\mathbb{Z}_p$	Closed and non-open

The quotient topologies are the usual ones on the displayed rings.

Proof. Using necklace coordinates, put

$$I_j = \{a \in R : a_1 = \cdots = a_j = 0\}.$$

These are ideals: every index contributing to a product coordinate $n \leq j$ divides n. They form a decreasing neighborhood basis of zero, and R is complete for this topology.

Let P be a closed prime and $D = R/P$. Since D is a domain, (6.1) implies that the image of e_n is either 0 or the image of the integer n. Every element of R is congruent modulo I_j to a finite integer combination of the e_n. Consequently the integer subring surjects onto $R/(P + I_j)$, and there is a unique $d_j \geq 0$ such that

$$R/(P + I_j) \cong \mathbb{Z}/d_j\mathbb{Z}, \qquad d_j \mid d_{j+1}, \tag{6.4}$$

where $\mathbb{Z}/0\mathbb{Z} = \mathbb{Z}$ and the transition maps preserve the integer classes.

The natural map from D to the inverse limit of (6.4) is injective because $\bigcap_j (P + I_j) = P$. It is also surjective. Given compatible classes, choose representatives $x_j \in R$. Once x_j has been

chosen, compatibility allows the next representative to be changed by an element of P so that $x_{j+1} - x_j \in I_j$. The resulting sequence converges coordinatewise, and its limit represents all the classes. The kernels of the coordinate maps on both sides are their neighborhood bases. Thus

$$D \cong \varprojlim_{j} \mathbb{Z}/d_j\mathbb{Z} \tag{6.5}$$

as topological rings.

If some $d_j = 0$, every later d_j is zero and $D \cong \mathbb{Z}$ discretely. Otherwise put $\alpha_\ell = \sup_j v_\ell(d_j)$ for each rational prime ℓ. The Chinese remainder theorem at each finite stage gives

$$D \cong \prod_{\ell:\alpha_\ell>0} C_\ell, \qquad C_\ell = \begin{cases} \mathbb{Z}/\ell^{\alpha_\ell}\mathbb{Z}, & \alpha_\ell < \infty, \\ \mathbb{Z}_\ell, & \alpha_\ell = \infty. \end{cases} \tag{6.6}$$

This description remains valid when infinitely many primes occur: a class modulo any particular d_j uses only finitely many of the factors. Since D is a nonzero domain, exactly one factor occurs, and a finite factor must have exponent one. Hence the only possibilities are $\mathbb{Z}$, $\mathbb{F}_p$, and $\mathbb{Z}_p$, with their usual topologies.

It remains to determine the quotient map $\eta : R \twoheadrightarrow D$. In characteristic zero, write $\eta(e_n) = n\varepsilon_n$, where $\varepsilon_n \in \{0, 1\}$. Applying η to (6.1) and cancelling ij gives

$$\varepsilon_{\operatorname{lcm}(i,j)} = \varepsilon_i\varepsilon_j. \tag{6.7}$$

Thus the set $S = \{n : \varepsilon_n = 1\}$ contains 1 and is closed under divisors and least common multiples. If $D = \mathbb{Z}$, continuity to a discrete ring makes S finite. Taking $m = \operatorname{lcm} S$, these closure properties say that S is exactly the divisor set of m. Therefore η and g_m agree on every e_n, hence on the dense finite-support subring, and continuity gives $\eta = g_m$.

If $D = \mathbb{Z}_p$, continuity implies that, for each r, all sufficiently large $n \in S$ are divisible by p^r. In particular, there are only finitely many p-free members of S. Let m be their least common multiple; it belongs to S and is prime to p. Divisor closure now bounds every non-p exponent in S by the corresponding exponent of m. The p-exponents must be unbounded. Otherwise S would be finite, and η would agree, by density, with the finite sum $\sum_{n\in S} na_n$, contradicting surjectivity onto $\mathbb{Z}_p$. Divisor and least-common-multiple closure consequently give

$$S = \{dp^k : d \mid m,\ k \geq 0\}.$$

The maps η and $L_{m,p}$ agree on all e_n, and continuity again gives equality.

Finally suppose $D = \mathbb{F}_p$. When $p \mid n$, (6.1) forces $\eta(e_n) = 0$. For $p \nmid n$, write $\eta(e_n) = n\varepsilon_n$. Equation (6.7) holds for p-free indices. The surviving set is finite because the quotient is discrete, and its divisor and least-common-multiple closure make it the divisor set of a p-free integer m. It follows that $\eta = g_m \bmod p$.

All the listed maps are surjective. For g_m and its reduction modulo p, use multiples of e_1. For $L_{m,p}$, place the base-p digits of a prescribed p-adic integer in the coordinates a_{p^k}, with all other coordinates zero. Their kernels are therefore prime. The first two kernels are open; the third is closed by continuity and is non-open because arbitrarily far-out vectors e_{p^k} have nonzero image p^k.

The parameters can be recovered from the kernels. For A_m, the vectors e_n outside the kernel record the divisor set of m. For $B_{m,p}$, they record the set $\{dp^k : d \mid m,\ k \geq 0\}$, which determines both m and p. For $M_{m,p}$, intersection with the integer subring determines p, and the surviving

p-free vectors determine m. The positive-characteristic family is disjoint from the other two, and the characteristic-zero families are separated by openness. □

The quotient rings also determine every prime above a characteristic-zero closed prime, without an additional closedness assumption.

Corollary 6.2 (Specializations of closed primes). The only strict containments between distinct closed primes are

$$A_m \subsetneq M_{m/p^{v_p(m)},p} \quad (p \text{ prime}), \qquad B_{m,p} \subsetneq M_{m,p}. \tag{6.8}$$

Every prime strictly above A_m or $B_{m,p}$ appears here.

Proof. Apply prime correspondence to the quotients $\mathbb{Z}$ and $\mathbb{Z}_p$. The pullback of $p\mathbb{Z}$ under g_m is the displayed M-ideal because $g_m \equiv g_{m/p^{v_p(m)}} \pmod p$. The pullback of $p\mathbb{Z}_p$ under $L_{m,p}$ is $M_{m,p}$. Each M-ideal is maximal. □

The statement concerns primes above the closed families. A nonclosed prime can lie strictly below one of them. For example, let $\mathcal{U}$ be a nonprincipal ultrafilter on the rational primes and put

$$P_{\mathcal{U}} = \{a \in R : \{q \text{ prime} : g_q(a) = 0\} \in \mathcal{U}\}.$$

This is prime as the kernel of a map into an ultraproduct of copies of the domain $\mathbb{Z}$. In necklace coordinates, $g_q(a) = a_1 + qa_q$. If this vanishes for an ultrafilter-large set of primes, infinitely many primes divide a_1, so $a_1 = 0$. Hence $P_{\mathcal{U}} \subseteq A_1$. The inclusion is strict: the necklace vector with value one at every prime index and zero elsewhere belongs to A_1, but its q-th ghost is q, so it is outside $P_{\mathcal{U}}$. Every finitely supported necklace vector with first coordinate zero belongs to $P_{\mathcal{U}}$. These vectors are dense in A_1, and therefore $\overline{P_{\mathcal{U}}}^{\tau} = A_1$. Thus a closed prime need not be minimal in the ordinary spectrum, and membership in an ordinary ideal cannot be inferred from membership of all terms of a convergent necklace expansion.

6.2. Extended ideals of the closed primes

A prime $Q \subset B$ contracting to $P \subset R$ must contain PB. To use this condition, we need the extended ideal itself, rather than its coordinate images alone: coordinatewise generation need not imply generation by a finite sum in a product ring. The two-ghost realization theorem supplies that finite generation here.

For $m \geq 1$, define

$$h_m(n) = \begin{cases} 0, & n = m, \\ q^{\min(v_q(n),v_q(m))+1}, & n/m = q^j,\ q \text{ prime},\ j \in \mathbb{Z} \setminus \{0\}, \\ 1, & \text{otherwise.} \end{cases} \tag{6.9}$$

The second case includes both increases and decreases of one prime exponent. For $p \nmid m$, define two further profiles by

$$h_{m,p}(n) = \begin{cases} p^{k+1}, & n = mp^k,\ k \geq 0, \\ 1, & \text{otherwise,} \end{cases} \qquad h^M_{m,p}(n) = \begin{cases} p, & n = mp^k,\ k \geq 0, \\ 1, & \text{otherwise.} \end{cases} \tag{6.10}$$

Proposition 6.3 (Extended ideals). In B,

$$A_m B = h_m B, \qquad B_{m,p} B = h_{m,p} B, \qquad M_{m,p} B = h^M_{m,p} B. \tag{6.11}$$

Each closed prime of R is the radical of a two-generated ideal contained in it.

Proof. The elements $e_d - g_m(e_d)e_1$ belong to A_m, with n-th ghosts

$$d(\mathbf{1}_{d|n} - \mathbf{1}_{d|m}). \tag{6.12}$$

They generate the integer ideal $g_n(A_m)$: for an arbitrary member of A_m, subtracting its zero m-th ghost expresses its n-th ghost as a finite integer combination of (6.12). If m and n differ at a single prime exponent q, the divisors occurring in their symmetric difference are all divisible by $q^{\min(v_q(n),v_q(m))+1}$, and that prime power itself occurs. If they differ at two or more prime exponents, the symmetric difference contains two coprime prime powers. Thus $g_n(A_m) = h_m(n)\mathbb{Z}$.

The profile h_m is compatible. Indeed, it is the coordinatewise gcd of the ghosts of all elements of A_m; at each prime, truncation of valuations at a fixed finite threshold commutes with taking their minimum. The two-ghost realization theorem therefore gives $f, g \in R$ with ghost gcd h_m. Both have zero m-th ghost, so $f, g \in A_m$. Coordinatewise Bézout identities in B give $(f, g)B = h_m B$, while every ghost from A_m is divisible by h_m. This proves the first equality in (6.11).

For the second, the tail of (6.2) gives

$$L_{m,p}(a) - g_{mp^k}(a) \in p^{k+1}\mathbb{Z}_p. \tag{6.13}$$

Every ghost from $B_{m,p}$ is consequently divisible by $h_{m,p}$. To check compatibility of this profile, consider an edge joining n and n/q, where $q \mid n$. For $q \neq p$, both q-valuations are zero. For $q = p$, either both indices lie outside $\{mp^k : k \geq 0\}$, or they are mp^k, mp^{k-1}, whose profile valuations $k+1, k$ agree after truncation at k. Hence two ghosts realize $h_{m,p}$. Their values at mp^k tend to zero p-adically, so their preimages belong to $B_{m,p}$, proving the second equality.

For the third, $M_{m,p} = A_m + pR$: if $g_m(a) = pz$, then $a - pze_1 \in A_m$. Therefore

$$M_{m,p} B = (h_m, p)B = \gcd(h_m, p)B = h^M_{m,p} B.$$

The last equality uses $p \nmid m$. The profile $h^M_{m,p}$ is compatible as the gcd of h_m and the constant profile p, and any two ghost realizations have m-th ghosts divisible by p, hence belong to $M_{m,p}$.

In each case we have found $f, g \in P$ with $(f, g)B = PB$. Radical contraction from Section 3 now gives

$$\sqrt{(f, g)R} = \sqrt{(f, g)B} \cap R = \sqrt{PB} \cap R = P.$$

This proves the last assertion as well. □

For $m = 1$, the middle profile is the running example $h(p^k) = p^{k+1}$, with value 1 elsewhere. We can now identify the ideal behind that example: $hB = B_{1,p}B$. Its unbounded exponents explain why the constant ghost p fails the uniform divisibility test, consistently with $p \notin \ker L_{1,p}$. The same exponents will distinguish the ambient primes in this fiber.

6.3. Coordinate ultrafilters

The idempotents of B separate the coordinate choices made by a prime. If $S \subseteq \mathbb{N}_{>0}$, let $\mathbf{1}_S$ be its characteristic sequence. For a prime $Q \subset B$, exactly one of $\mathbf{1}_S, \mathbf{1}_{S^c}$ lies in Q. The collection

$$\mathcal{F}_Q = \{S : \mathbf{1}_S \notin Q\} \tag{6.14}$$

is an ultrafilter: it is closed under finite intersections and passage to supersets, excludes the empty set, and chooses one member of every complementary pair. A principal ultrafilter chooses one coordinate; a nonprincipal ultrafilter on our countable index sets contains every cofinite subset.

The ultraproduct $D_{\mathcal{F}} = \prod_{\mathcal{F}} \mathbb{Z}$ identifies two integer sequences when they agree on a set belonging to $\mathcal{F}$. It is a domain, since a zero product has a zero factor on an ultrafilter-large set. If a sequence is zero on such a set S, it is a multiple of $\mathbf{1}_{S^c}$, which belongs to Q. Thus the kernel of $B \to D_{\mathcal{F}_Q}$ lies in Q, and Q corresponds to a prime of this domain.

This is the coordinate-ultrafilter part of the established product-spectrum descriptions [11; 12, Theorem 3.6]. We will prove the valuation description needed below directly. We also record an immediate consequence of (6.14): primes with different coordinate ultrafilters are incomparable, because a set selected by one ultrafilter and rejected by the other gives complementary idempotents separating the two primes.

6.4. The fibers over A_m and $M_{m,p}$

For A_m, the profile (6.9) singles out the coordinate m, together with sequences obtained by changing one prime exponent. The characteristic-zero condition will eliminate every fixed prime label from the latter possibility. Put

$$C_m = \{mq^j : q \text{ prime},\ q \nmid m,\ j \geq 1\},$$

and let $q(n)$ denote the unique prime label of $n \in C_m$.

Theorem 6.4 (The fiber over A_m). The primes of B contracting to A_m are the coordinate prime

$$Z_m = \{x \in B : x_m = 0\} \tag{6.15}$$

and the maximal ideals

$$Q_{m,\mathcal{F}} = \{x \in B : \{n \in C_m : q(n) \mid x_n\} \in \mathcal{F}\}, \tag{6.16}$$

where $\mathcal{F}$ ranges over ultrafilters on C_m satisfying

$$\{n \in C_m : q(n) = q_0\} \notin \mathcal{F} \quad \text{for every fixed prime } q_0. \tag{6.17}$$

Proof. Let $Q \cap R = A_m$. Then $h_m \in Q$ by Proposition 6.3, and $Q \cap \mathbb{Z} = 0$. If its coordinate ultrafilter is principal at n, the corresponding prime of $\mathbb{Z}$ must be zero. Hence $h_m(n) = 0$, forcing $n = m$ and $Q = Z_m$.

For a nonprincipal ultrafilter, we may discard the singleton $\{m\}$; the value $h_m = 1$ outside the single-prime branches then forces concentration on those branches. It cannot select the branch for a fixed prime q: there h_m divides the fixed integer $q^{v_q(m)+1}$, so $h_m \in Q$ would imply $q \in Q$.

We may therefore discard the finitely many branches for primes dividing m. What remains is C_m, where $h_m(n) = q(n)$, and the induced ultrafilter satisfies (6.17).

In $D_{\mathcal{F}}$, the ideal generated by $[q(n)]_{\mathcal{F}}$ is maximal, because its quotient is $\prod_{\mathcal{F}} \mathbb{F}_{q(n)}$. This ultraproduct is a field: a nonzero class has nonzero coordinates on an ultrafilter-large set, where its coordinatewise inverses define an inverse class. Any prime containing $[q(n)]_{\mathcal{F}}$ is therefore its principal ideal, giving (6.16).

Conversely, successive ghost congruences along $m, mq, \dots, mq^j$ give

$$g_{mq^j}(a) \equiv g_m(a) \pmod{q}.$$

If $g_m(a) = 0$, membership in (6.16) follows. If $g_m(a) \neq 0$, only finitely many prime labels divide that integer, and (6.17) excludes their union from $\mathcal{F}$. Thus $Q_{m,\mathcal{F}} \cap R = A_m$. The assertion for Z_m follows from its definition. □

The positive-characteristic closed fiber uses the same p-power sequence throughout, with a constant exponent.

Proposition 6.5 (The fiber over $M_{m,p}$). For $p \nmid m$, all primes over $M_{m,p}$ are

$$T_{m,p,\mathcal{F}} = \{x \in B : \{k \geq 0 : p \mid x_{mp^k}\} \in \mathcal{F}\}, \tag{6.18}$$

where $\mathcal{F}$ is any ultrafilter on $\mathbb{N}_0$. They are maximal and have residue field $\mathbb{F}_p$.

Proof. Proposition 6.3 gives

$$B/M_{m,p}B \cong \prod_{k\geq 0} \mathbb{F}_p.$$

Each prime of this product selects a coordinate ultrafilter, and its quotient ultraproduct is $\mathbb{F}_p$: a sequence in a finite field takes one value on an ultrafilter-large set. The ultrafilter kernel is therefore already maximal, proving that all the primes are (6.18). Their contractions are proper ideals containing the maximal ideal $M_{m,p}$, so all contract to it. □

6.5. The fiber over $B_{m,p}$

The profile $h_{m,p}$ also confines the fiber to $\{mp^k : k \geq 0\}$, but its increasing exponents leave room for different tests of divisibility. To express those tests, fix a nonprincipal ultrafilter $\mathcal{F}$ on $\mathbb{N}_0$ and form the ordered abelian group

$$\Gamma_{\mathcal{F}} = \prod_{\mathcal{F}} \mathbb{Z}, \qquad \kappa = [k+1]_{\mathcal{F}}. \tag{6.19}$$

Order is determined on an ultrafilter-large set. We identify $\mathbb{Z}$ with the subgroup of constant classes; κ exceeds every such positive integer. A subgroup $H \subseteq \Gamma_{\mathcal{F}}$ is convex if $0 \leq \gamma \leq h$ with $h \in H$ implies $\gamma \in H$.

For $x \in B$, take the coordinate p-valuations along the selected sequence:

$$\nu_{\mathcal{F},p}(x) = [v_p(x_{mp^k})]_{\mathcal{F}}. \tag{6.20}$$

If $x_{mp^k} = 0$ on a set in $\mathcal{F}$, set $\nu_{\mathcal{F},p}(x) = \infty$. Otherwise the zero set of x_{mp^k} can be discarded and the finite valuations define a nonnegative element of $\Gamma_{\mathcal{F}}$. We order ∞ above the group.

Theorem 6.6 (The p-adic fiber). All primes of B contracting to $B_{m,p}$ are

$$Q_{m,p,\mathcal{F},H} = \{x \in B : \nu_{\mathcal{F},p}(x) > h \text{ for every } h \in H_{\geq 0}\}, \tag{6.21}$$

where $\mathcal{F}$ is nonprincipal and H is a convex subgroup satisfying

$$\mathbb{Z} \subseteq H, \qquad \kappa \notin H. \tag{6.22}$$

Proof. Suppose $Q \cap R = B_{m,p}$. Since $h_{m,p} \in Q$ and equals 1 off the sequence mp^k, the coordinate ultrafilter concentrates on that sequence. It cannot be principal, because at a fixed coordinate the inclusion $p^{k+1} \in Q$ would force $p \in Q$, contradicting characteristic zero. Write $D = \prod_{\mathcal{F}} \mathbb{Z}$ for the resulting domain. The corresponding prime $\mathfrak{q}$ contains

$$p^{\kappa} = [p^{k+1}]_{\mathcal{F}}.$$

Every such prime lies inside pD. Indeed, if $u \notin pD$, then u_k is prime to p on a set in $\mathcal{F}$. Coordinatewise Bézout identities give $au + bp^{\kappa} = 1$ in D, so $u \notin \mathfrak{q}$. Also pD is maximal because $D/pD \cong \mathbb{F}_p$.

We can now use the prime correspondence for $V = D_{pD}$. This ring is a valuation domain with value group $\Gamma_{\mathcal{F}}$. On a nonzero fraction $a/b \in \operatorname{Frac} D$, define its value by the difference of the coordinate p-valuations. If this difference is nonnegative, cancel $p^{v_p(b_k)}$ from numerator and denominator on an ultrafilter-large set. The resulting numerator has integer coordinates and the denominator has valuation zero, so the fraction lies in D_{pD}. Conversely, every fraction in D_{pD} has nonnegative value. Every nonnegative value is realized by a sequence of powers of p. This proves the valuation-domain assertion and identifies its full value group.

For completeness, the prime ideals of a valuation domain correspond to convex subgroups of its value group as follows. For a prime $\mathfrak{r} \subset V$, let S be the values of its nonzero elements outside $\mathfrak{r}$. Membership depends only on value, since elements of equal value differ by a unit. The set S is closed under addition because the complement of a prime is multiplicatively closed. It is downward closed: if $0 \leq \beta \leq \alpha \in S$, an element of value α factors, up to a unit, into elements of values β and $\alpha - \beta$, and both factors are outside $\mathfrak{r}$. Thus $H = S - S$ is a convex subgroup with $H_{\geq 0} = S$. Conversely, the elements with value greater than every member of $H_{\geq 0}$, together with zero, form a prime ideal: the valuation inequality proves closure under addition, and a sum of two values in $H_{\geq 0}$ remains in $H_{\geq 0}$. This proves the correspondence used in (6.21).

Under that correspondence, $p^{\kappa} \in \mathfrak{q}$ means $\kappa \notin H$, while $p \notin \mathfrak{q}$ means $1 \in H$, equivalently $\mathbb{Z} \subseteq H$. Every other fixed rational prime is a unit of V. These are exactly the necessary conditions (6.22).

They are sufficient. The valuation construction gives an ambient prime. If $L_{m,p}(a) = 0$, (6.13) implies

$$v_p(g_{mp^k}(a)) \geq k + 1.$$

Its ultraproduct value is therefore at least κ, hence above every nonnegative member of H by convexity. If $L_{m,p}(a) \neq 0$, p-adic convergence gives

$$v_p(g_{mp^k}(a)) = v_p(L_{m,p}(a))$$

for all sufficiently large k. This is a fixed finite integer, which belongs to H. The contraction of (6.21) is consequently exactly $B_{m,p}$. □

Conditions (6.22) retain the two features of the running profile: fixed powers of p stay outside the prime, whereas the increasing power p^{k+1} lies inside. Different convex subgroups can distinguish other rates of growth, even though every resulting prime has the same contraction to R.

6.6. Equality and inclusion among the lifts

The ultrafilter of an ambient prime is recovered from its idempotents, so the descriptions above have no ambiguity in that parameter. The fiber over A_m is an antichain: all its points have different coordinate ultrafilters, and its nonprincipal points are maximal. The fiber over $M_{m,p}$ is an antichain of maximal ideals.

For a fixed $\mathcal{F}$ in the p-adic fiber, convex subgroups are linearly ordered and

$$Q_{m,p,\mathcal{F},H_1} \subseteq Q_{m,p,\mathcal{F},H_2} \quad \Longleftrightarrow \quad H_2 \subseteq H_1. \tag{6.23}$$

The forward implication can be tested on sequences of powers of p, which realize every nonnegative value; the reverse implication follows from (6.21). Equality therefore holds exactly when $H_1 = H_2$. Different ultrafilters give incomparable primes, so the p-adic fiber is a disjoint union of chains.

Finally, inclusions across different closed-prime fibers can be read from their coordinate ultrafilters and Corollary 6.2. The proper superprimes of Z_m are precisely the coordinate mod-p primes at m; they contract to $M_{m/p^{v_p(m)},p}$. The nonprincipal primes over A_m are maximal in B, although A_m itself has the specializations in (6.8). For $Q_{m,p,\mathcal{F},H}$, the unique maximal superprime is $T_{m,p,\mathcal{F}}$. Indeed, every superprime retains the same coordinate ultrafilter and contains p^κ, so the Bézout argument in Theorem 6.6 places it inside pD. The subgroup rule (6.23) describes the intermediate superprimes in the $B_{m,p}$-fiber. The group $\Gamma_{\mathcal{F}}$ has least positive element 1, so every nonzero convex subgroup contains $\mathbb{Z}$; the only smaller subgroup that leaves this fiber is $\{0\}$, corresponding to $T_{m,p,\mathcal{F}}$. Contraction preserves inclusion, and Corollary 6.2 leaves no further possibilities between distinct closed-prime fibers.

7. Further questions

The arithmetic presentation separates two tasks that the ghost embedding alone does not settle. The uniform square bound allows every prime to be lifted to the product. Compatible profiles then determine which of those ambient distinctions survive contraction. For the coordinate-topology-closed primes, this leads to the explicit fibers of Section 6. For arbitrary primes, the signature and normalized-profile criteria remain infinitary.

A first question is whether the bound of two on arithmetical rank is optimal for this ring. Theorem 4.2 realizes every compatible profile by two ghosts, but a profile need not itself be a ghost. This does not decide whether its uniform-power equivalence class always contains the absolute value of a single ghost. Such a result would make every finitely generated ideal have the same radical as a principal ideal; a counterexample would show that the bound two is attained. The realization theorem supplies a numerical formulation of the question, without deciding it.

A second question concerns the ambient description away from the closed families. The profile criteria decide equality and inclusion of contractions, but one can ask for simpler conditions directly on ultrafilters and valuation subgroups that characterize the minimal and maximal contractions. Any such description must preserve the uniform power comparisons that distinguish radicals in (3.11). The closed fibers provide examples against which a proposed simplification can be tested.

A third question is which features of ideal transfer persist for other coefficient rings or other profinite groups. In the present proof, the finite Bézout identity and the local predecessor congruences cooperate at each coordinate. Changing either the coefficients or the orbit indexing

changes that construction. A useful extension would therefore identify explicit arithmetic hypotheses under which a uniform bound $(IC \cap A)^s \subseteq I$ holds for a ghost embedding $A \subseteq C$, and determine how the exponent depends on those hypotheses. No such general bound is asserted here.

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Appendix A. Propagation and forced zeros in the normalizer

We prove the formula (5.8). Its construction has two parts because divisibility can force a zero entry either from an existing zero or from infinitely many distinct prime factors.

Starting with $Z_0 = \{n : d_n = 0\}$, define

$$Z_{\gamma+1} = Z_\gamma \cup \left\{ n : \begin{array}{l} \text{for infinitely many primes } p \nmid n, \text{ some } j \geq 1 \text{ satisfies} \\ p \mid d_{np^j} \text{ or } np^j \in Z_\gamma \end{array} \right\}. \tag{A.1}$$

At a limit ordinal take the union. This increasing sequence stabilizes at a countable ordinal: otherwise every countable successor stage would add a new positive integer, giving uncountably many distinct integers. Its stable value is exactly the least set Z specified in (5.6). This is a transfinite closure construction; no finite termination assertion is needed.

Any compatible multiple e of d vanishes on Z. It vanishes first on Z_0. Suppose it vanishes on Z_γ and that n is added at the next stage. For each prime branch in (A.1), either divisibility by d_{np^j} or the zero value at np^j gives $p \mid e_{np^j}$. Applying (4.1) successively along the p-chain gives $p \mid e_n$. Since this holds for infinitely many distinct primes, $e_n = 0$. The assertion passes through limit stages by taking unions.

Fix p, write $n = mp^k$ with $p \nmid m$, and abbreviate $\alpha_j = \alpha(mp^j, p)$. A sequence of valuations $x_k \in \mathbb{N}_0 \cup \{\infty\}$ satisfies compatibility on this chain precisely when

$$\min(x_k, k) = \min(x_{k-1}, k) \qquad (k \geq 1). \tag{A.2}$$

If $x_j \geq \alpha_j$, propagation across the edges from j to k gives

$$x_k \geq \min\{\alpha_j, c(k, j)\}. \tag{A.3}$$

Indeed, each edge at level r transfers a lower bound up to r, and the least level on a nonempty path from j to k is $\min(k, j) + 1$. At $j = k$ no cap is imposed. Every solution above the initial bounds therefore dominates $\beta_k = \sup_j \min\{\alpha_j, c(k, j)\}$.

These β_k themselves satisfy (A.2). For every $k \geq 1$ and every $j \geq 0$,

$$\min\{c(k, j), k\} = \min\{c(k-1, j), k\}. \tag{A.4}$$

Truncating the definition of β_k at k, commuting the finite truncation with the supremum, and using (A.4) gives $\min(\beta_k, k) = \min(\beta_{k-1}, k)$. The term $j = k$ also gives $\beta_k \geq \alpha_k$. Hence (5.7) is the least compatible valuation sequence above the prescribed bounds on each p-chain.

It remains to check that these separate valuation sequences define finite integers outside Z. If $n = mp^k \notin Z$, the term $j = k$ in (5.7) is finite, and every other term is bounded by $k + 1$. Thus $\beta(n, p) < \infty$. Only finitely many primes divide nd_n. For any remaining prime p, we have $k = 0$, and positivity of $\beta(n, p)$ requires some $j \geq 1$ with $p \mid d_{np^j}$ or $np^j \in Z$. If this occurred for infinitely many primes, the fixed-point rule (5.6) would put n in Z. Therefore only finitely many $\beta(n, p)$ are positive.

Formula (5.8) consequently defines an integer profile. At $n \in Z$, the self-term in (5.7) makes every valuation infinite, agreeing with the assigned zero entry. At other indices its valuations are

the finite $\beta(n,p)$. The chain calculation proves compatibility, and the self-terms prove divisibility by d. Finally, every compatible multiple of d vanishes on Z, hence has valuations at least $\alpha(n,p)$; equation (A.3) then forces valuations at least $\beta(n,p)$. Thus the profile in (5.8) divides every compatible multiple of d, and is $\sigma(d)$. □

Appendix B. Ambient valuation parameters and profile signatures

The point model in Theorem 5.4 does not require a choice of ambient prime. This appendix gives the translation for readers using the product-spectrum parameters of [11, 12]. The ultraproduct and convex-subgroup conventions are those explained in Sections 6.3 and 6.5.

Let $Y = \mathbb{N}_{>0} \times \mathcal{P}$, where $\mathcal{P}$ is the set of rational primes. For an ultrafilter $\mathcal{U}$ on Y, put $\Gamma_{\mathcal{U}} = \mathbb{Z}^Y/\mathcal{U}$, ordered by comparison on an ultrafilter-large set, and let H be a convex subgroup. For $b \in B$, define

$$v_{\mathcal{U}}(b) = [v_p(b_n)]_{\mathcal{U}}.$$

If $\{(n,p) \in Y : b_n = 0\} \in \mathcal{U}$, this value is ∞. Otherwise discard that zero set, take the finite valuations on its complement, and fill the discarded entries arbitrarily. The resulting group element is independent of the filling. As before, ∞ is greater than every group element.

Each set

$$Q_{\mathcal{U},H} = \{b \in B : v_{\mathcal{U}}(b) > h \text{ for every } h \in H_{\geq 0}\} \tag{B.1}$$

is prime. Indeed, map B to the valuation domain $E = \prod_{(n,p)\in Y} \mathbb{Z}_{(p)}/\mathcal{U}$ by repeating b_n at each pair (n,p). Its value group is $\Gamma_{\mathcal{U}}$, since arbitrary nonnegative values are realized by coordinate-wise powers of p. The convex-subgroup correspondence proved in Section 6.5 gives a prime of E, and (B.1) is its inverse image.

These parameters exhaust $\operatorname{Spec} B$. To specify the dependence on the established classification, let Q be any ambient prime. By [12, Propositions 2.30–2.31], choose $\mathcal{U}$ with

$$N_{\mathcal{F}} \subseteq Q \subseteq M_{\mathcal{U}},$$

where $\mathcal{F}$ is the coordinate projection of $\mathcal{U}$,

$$N_{\mathcal{F}} = \{b : \{n : b_n = 0\} \in \mathcal{F}\},$$
$$M_{\mathcal{U}} = \{b : \{(n,p) : p \mid b_n\} \in \mathcal{U}\},$$

and $M_{\mathcal{U}}$ is maximal. The prime Q corresponds to a prime of

$$V = (B/N_{\mathcal{F}})_{M_{\mathcal{U}}/N_{\mathcal{F}}}.$$

By [12, Proposition 3.3], V is the valuation ring of the restriction of $v_{\mathcal{U}}$, with realized value group $G_{\mathcal{U}} \subseteq \Gamma_{\mathcal{U}}$. Let H_0 be the convex subgroup of $G_{\mathcal{U}}$ corresponding to that prime, and put

$$H = \{\gamma \in \Gamma_{\mathcal{U}} : |\gamma| \leq h \text{ for some } h \in (H_0)_{\geq 0}\}.$$

This is a convex subgroup of $\Gamma_{\mathcal{U}}$, and $H \cap G_{\mathcal{U}} = H_0$ by convexity in the realized group. Formula (B.1) therefore gives Q. This reformulates the localization description using convex subgroups; it does not assume that the restricted valuation has image all of $\Gamma_{\mathcal{U}}$.

Since $J_d B = dB$, the signature of the contraction of (B.1) is

$$\varepsilon_{\mathcal{U},H}(d) = 1 \quad \Longleftrightarrow \quad [v_p(d_n)]_{\mathcal{U}} > h \quad \text{for every } h \in H_{\geq 0}. \tag{B.2}$$

Every signature arises in this way by Corollary 3.2. Different pairs $(\mathcal{U}, H)$ may give the same signature; Theorem 5.2 gives equality and containment tests for their contractions. The endpoint conventions in (B.1) are explicit: $H = \{0\}$ tests positive valuation, and $H = \Gamma_{\mathcal{U}}$ tests vanishing on an ultrafilter-large set. The signature description retains the distinctions visible to R without choosing a unique ambient representative.

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